Yingyi Zhang

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Education Background

City University of Hong Kong (CityU) — *Hong Kong*
Joint PhD student in Data Science, supervised by Prof. Xiangyu Zhao — Sep. 2024 - Now

Dalian University of Technology (DUT) — *Dalian, China*
PhD student in Management Science and Engineering, supervised by Prof. Xianneng Li — Sep. 2020 - Now

Dalian University of Technology (DUT) — *Dalian, China*
BS in Information Management and Information System — Sep. 2016 - Jun. 2020

Brief Biography

Yingyi Zhang is a joint Ph.D. candidate at Dalian University of Technology and City University of Hong Kong. His research focuses on personalized RAG for LLM and personalized consumer behavior understanding in RecSys.

- **Research Output:** Published several papers in top venues, including *ICLR, ICML, NeurIPS, KDD, WWW, AAAI, ACL, ACM TOIS, ACM CSUR* and *Journal of Management Sciences in China*.
- **Competition Output:** KDD Cup: ranked 2nd in the User Behavior Alignment track (1st: NVIDIA) and received the *Second Prize and Student Award*. WWW Competition 2025 (Multimodal Dialogue Intent Recognition): finalist & awarded a Comprehensive Bronze, with *1st-Place Final-Stage Ranking*.
- **2026.4-Now: ByteDance (Frontier Tech Project)** Deployed Agentic Fusion Tuner systems in Qishui Music recommendation, delivering *+0.09% LT14 uplift per day, +0.18% playtime per user, and +0.89% playtime per pv*, with related work submitted to *KDD'27*.
- **2024.9-2026.4: Huawei** Proposed a cloud–device LLM–SLM collaboration framework, personalized pre-retrieval pipeline, and personalized retrieval pipeline. Three papers were accepted to *KDD'25*, *AAAI'26*, and *ICLR'26*.
- **2021.9-2022.7: Meituan** Deployed recommender systems in Meituan Search, delivering *+3.14% GTV uplift per day (1.59%+1.55%) and +934 daily UV*, with related work published at *WWW'23* and *DL4SR@CIKM'22*.

Competition

WWW Competition 2025: Multimodal Dialogue System Intent Recognition Challenge — *Online*
Field of E-commerce Customer Service and LLM [Link] — Nov. 2024 - Jan. 2025

- *1st place in the final round, and bronze medal rated by the experts*
- Multimodal Intent Recognition in Customer Service via Data Augment and LLM Merge

KDD Cup 2024: A Multi-task Online Shopping Challenge for Large Language Models — *Online*
Field of Recommender Systems and LLM [Link] — May. 2024 - Jul. 2024

- *2nd place in the User Behavior Alignment Track with 1st is Team NVIDIA*
- Enhancing User Behavior Alignment by Input-Level Model Cooperation and Model-Level Parameter Optimization

Publications

First and co-first authors:

[1] **Yingyi Zhang**†, Junyi Li†,..., Xianneng Li*, Xiangyu Zhao*, et al. 2026. Evoking User Memory: Personalizing LLM via Recollection-Familiarity Adaptive Retrieval. [Link] *International Conference on Learning Representations. ICLR'26*

[2] **Yingyi Zhang**†, Pengyue Jia†,..., Xianneng Li*, Xiangyu Zhao*, et al. 2026. Personalize Before Retrieve: LLM-based Personalized Query Expansion for User-Centric Retrieval. *Accepted by AAAI 26 as an oral paper.* [Link] *AAAI'26-Oral*

[3] **Yingyi Zhang**†, Pengyue Jia†,..., Xianneng Li*, Xiangyu Zhao*, et al. 2025. LSRP: A Leader–Subordinate Retrieval Framework for Privacy-Preserving Cloud–Device Collaboration. *In Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2, August 3-7, 2025, Toronto, ON, Canada.* [Link] *KDD'25*

[4] **Yingyi Zhang**†, Zhipeng Li†, Zhewei Zhi†, and Xianneng Li*. 2024. Enhancing User Behavior Alignment by Input-Level Model Cooperation and Model-Level Parameter Optimization. *Amazon KDD Cup 2024 Workshop, August 25–29, 2024, Barcelona, Spain.* [Link] *Amazon KDD Cup'24*

[5] **Yingyi Zhang**, Xianneng Li*, Yanhong Guo, et al. 2024. Deep review-based recommendation from the perspective of consumer decision journey[J]. *Journal of Management Sciences in China. Top 1 Chinese Journal in Management and Science*

[6] **Yingyi Zhang**, Xianneng Li*, Yahe Yu, et al. 2023. Meta-Generator Enhanced Multi-Domain Recommendation.*In Companion Proceedings of the ACM Web Conference 2023 (WWW'23 Companion - industry truck), April 30-May 4, 2023, Austin, TX, USA.*[Link] *WWW'23-Industry, with an Online Deployed Systems in Meituan*

[7] **Yingyi Zhang**, Xianneng Li*, Yahe Yu, et al. 2022. Entire Cost Enhanced Multi-Task Model for Online-to-Offline Conversion Rate Prediction. *DL4SR'22: Workshop on Deep Learning for Search and Recommendation, co-located with the 31st ACM International Conference on Information and Knowledge Management (CIKM), October 17-21, 2022, Atlanta, USA.* [Link] *DL4SR'22, with an Online Deployed Systems in Meituan*

[8] Pengyue Jia†, **Yingyi Zhang**†, Xiangyu Zhao, Sharon Li. 2026. GeoArena: An Open Platform for Benchmarking Large Vision-language Models on WorldWide Image Geolocalization *Accepted by ACL'26. ACL'26-Main*

[9] Maolin Wang†, **Yingyi Zhang**†, Bowen Yu†,..., Xiangyu Zhao*, et al. 2026. Function Calling in Large Language Models: Industrial Practices, Challenges, and Future Directions. *Accepted by ACM SCUR. Top 1 Survey Journal in Computer Science, IF=26.3*

[10] Zhewei Zhi†, **Yingyi Zhang**†, Xianneng Li*, et al. 2026. Compress-then-Rank: Faster and Better Listwise Reranking with Large Language Models via Ranking-Aware Passage Compression. *Accepted by AAAI 26 as a poster paper. AAAI'26*

[11] Zhipeng Li†, Binglin Wu†, **Yingyi Zhang**†, Xianneng Li*, et al. 2025. CuSMer: Multimodal Intent Recognition in Customer Service via Data Augment and LLM Merge. *In Companion Proceedings of the ACM Web Conference 2025, April 28-May 2, 2025, Sydney, NSW, Australia.*[Link] *WWW'25-Compitition*

[12] Pengyue Jia†, Xiaobei Wang†, **Yingyi Zhang**†, Shuchang Liu, Yupeng Hou, Hailan Yang, Xu Gao, Xiaopeng Li, Yejing Wang, Julian McAuley, Xiang Li, Lantao Hu, Yongqi Liu, Kaiqiao Zhan, Han Li, Kun Gai and Xiangyu Zhao. GloRank: Generative Reranking for Deep Recommender Systems via Global Identifier Generation. *Accepted by CIKM 26. CIKM'26*

Co-authors:

[1] Junyi Li, Xiaowei Qian, **Yingyi Zhang**, Wenlin Zhang, Guojing Li, Sheng Zhang, Xiao Han, Yichao Wang and Xiangyu Zhao. 2026**Towards Pareto-Optimal Tool-Integrated Agents with Pareto Ranking Policy Optimization.** *Accepted by ICML'26. ICML'26-Spotlight*

[2] Sheng Zhang, Junyi Li, **Yingyi Zhang**, Pengyue Jia, Yichao Wang, Xiaowei Qian, Wenlin Zhang, Maolin Wang, Yong Liu abd Xiangyu Zhao. 2026. MemSearch-o1: Empowering Large Language Models with Reasoning-Aligned Memory Growth in Agentic Search.*Accepted by ACL'26. ACL'26-Main*

[3] Derong Xu, Shuochen Liu, Pengfei Luo, Pengyue Jia, **Yingyi Zhang**, Yi Wen, Yimin Deng, Wenlin Zhang, Enhong Chen, Xiangyu Zhao and Tong Xu. 2026. Learning How and What to Memorize: Cognition-Inspired Two-Stage Optimization for Evolving Memory. *Accepted by ACL'26. ACL'26-Main*

[4] Pengyue Jia, Xiaopeng Li, Derong Xu, Yi Wen, **Yingyi Zhang**, Wenlin Zhang, Wanyu Wang, Yichao Wang, Yong Liu, Huifeng Guo and Ruiming Tang. Bridging Personalization and AI: From RAG to Agent. *Accepted by SIGIR'26. SIGIR'26-Tutorial*

[5] Xiaopeng Li, Wenlin Zhang, **Yingyi Zhang**, Pengyue Jia, Yejing Wang, Yichao Wang, Yong Liu, Huifeng Guo and Xiangyu Zhao. Personalized Deep Research: A User-Centric Framework, Dataset, and Hybrid Evaluation for Knowledge Discovery. Accepted at SIGIR'26 Resource track. *Accepted by SIGIR'26. SIGIR'26-Resource*

[6] Wenlin Zhang, Kuicai Dong, Junyi Li, **Yingyi Zhang**, Xiaopeng Li, Pengyue Jia, Yi Wen, Derong Xu, Maolin Wang, Yichao Wang, Yong Liu and Xiangyu Zhao. 2026. To Search or Not to Search: Aligning the Decision Boundary of Deep Search Agents via Causal Intervention, *Accepted by WWW'26. WWW'26-Research*

[7] Xiaopeng Li, Pengyue Jia, Derong Xu, Yi Wen, **Yingyi Zhang**, Wenlin Zhang, Wanyu Wang, Yichao Wang, Xiangyang Li, Zhaocheng Du, Yong Liu, Huifeng Guo, Ruiming Tang, Xiangyu Zhao. 2026. A Survey of Personalization: From RAG to Agent,*Accepted by ACM TOIS. TOIS*

[8] Wenlin Zhang, Xiangyang Li, Kuicai Dong, Yichao Wang, Pengyue Jia, Xiaopeng Li, **Yingyi Zhang**, Derong Xu, Zhaocheng Du, Huifeng Guo, Ruiming Tang, Xiangyu Zhao. 2025. Process vs. Outcome Reward: Which is Better for Agentic RAG Reinforcement Learning,*In Proceedings of the 39th Conference on Neural Information Processing Systems.* [Link] *NeurIPS'25*

[9] Binglin Wu, **Yingyi Zhang**, Xianneng Li, Ruyue Deng, Chuan Yue, Weiru Zhang and Xiaoyi Zeng. PRO-Bid: Pareto-Prioritized Regret Optimization for Constraint-Aware Generative Auto-Bidding. *Accepted by CIKM 26. CIKM'26*

[10] Yue Que, **Yingyi Zhang**, Xiangyu Zhao*, and Chen Ma. 2025. Causality-aware Graph Aggregation Weight Estimator for Popularity Debiasing in Top-K Recommendation. *In Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM '25).* [Link] *CIKM'25*

[11] Zerong Lan, **Yingyi Zhang**, and Xianneng Li*. 2023. M3REC: A Meta-based Multi-scenario Multi-task Recommendation Framework, *In Proceedings of the 17th ACM Conference on Recommender Systems (RecSys '23).* 18th-22nd September 2023, Singapore. [Link] *RecSys'23*

[12] Yudi Xiao, **Yingyi Zhang** and Xianneng Li*. 2023. Modeling Variational Anchoring Effect for Recommender Systems,*In 2024 IEEE Conference on Artificial Intelligence (CAI) (pp. 926-931).* [Link] *IEEE CAI'23*

Recent Under Submission:

[1] **Yingyi Zhang**, Junyi Li, Yejing Wang, Wenlin Zhang, Xiaowei Qian, Sheng Zhang, Yue Feng, Yichao Wang, Yong Liu, Xiangyu Zhao, Xianneng Li. RAGR: Review-Augmented Generative Recommendation. *Submit to TOIS.*

[2] **Yingyi Zhang**, Chao Zhang, Junyi Li, Pengyue Jia, Jiancheng Dong, Haotian Nan, Wenlin Zhang, Ziwei Liu, Tongzhou Wu, Yuxuan Hu, Wanyu Wang, Mingfei Han, He Zhu, yansu, Yitan Li, Xianneng Li, Xiangyu Zhao. One Size Fits None: Agentic Fusion-Weight Tuning for Industrial Multi-Task Ranking. *Submit to KDD 2027.*

[3] Haotian Nan, **Yingyi Zhang**, Chao Zhang, Ziwei Liu, Yuxuan Hu, Jiancheng Dong, Tongzhou Wu, Xiaopeng Li, Derong Xu, Tianbo Huang, Lihua Zhang, Xiangyu Zhao. FISCHL: Fixed-Prefix Insertion of Semantically Retrieved Cold Items through History-Conditioned Local Search. *Submit to KDD 2027.*

[4] Wenlin Zhang, Jingtong Gao, Yu Rong, Zichuan Fu, Yewen Li, Pengyue Jia, Kaifeng Guo, **Yingyi Zhang**, Xiaopeng Li, Peng Jiang, Qingpeng Cai, Xiangyu Zhao. MARC: Scaling Agentic Multi-Channel Retrieval with Residual-Cohort Inference. *Submit to KDD 2027.*

[5] Binglin Wu, Chuan Yue, **Yingyi Zhang**, Xianneng Li, Ruyue Deng, Weiru Zhang, Xiaoyi Zeng. Beyond Single-Episode Optimization: Sliding-Window Aware Generative Auto-Bidding for Long-Term Advertising Effectiveness. *Submit to KDD 2027.*

[6] Yi Wen, Hao Chen, Wanyu Wang, Maolin Wang, Pengyue Jia, Derong Xu, Hui-Ze Tan, **Yingyi Zhang**, Wenlin Zhang, Weihong Luo, Xiku Du, Xiangyu Zhao. TrajEvolve: Trajectory Evolution for Reinforcement Learning with Hindsight Credit Assignment. *Submit to NeurIPS 2026.*

[7] Pengyue Jia, Derong Xu, **Yingyi Zhang**, Xiaopeng Li, Wenlin Zhang, Yi Wen, Lan Liu, Yuanshao Zhu, Xuetao Wei, Wanyu Wang, Xiangyu Zhao. GeoRouter: Dynamic Paradigm Routing for Worldwide Image Geolocalization. *Submit to NeurIPS 2026.*

Awards and Honors

Apr. 2026	**Volunteer Scholarship**: In ICLR 2026	*400 USD*
Jan. 2026	**Student Scholarship**: In AAAI 2026	*1,000 USD*
Apr. 2025	**Bronze medal**: WWW'25 Competition: Multimodal Dialogue System Intent Recognition Challenge	*7,000 CNY*
Sep. 2024	**Graduate Research Fund of the School of Economics and Management of DUT**	*50,000 CNY*
Aug. 2024	**Second Prize and Student Award**: KDD Cup 2024: Multi-Task Online Challenge for LLMs, User Behavior Alignment Track	*1,750 USD*
Oct. 2023	**National Third Prize**: Supreme People's Procuratorate National Procuratorate Big Data Legal Supervision	-
Sep. 2023	**Outstanding Graduate Student**: Dalian University of Technology	-
Oct. 2023	**Commended Paper**: The eleventh CNAIS Annual Conference.	-
Oct. 2021	**Best Paper**: The First Academic Conference on Data Intelligence and Management	*1,000 CNY*

Research Projects

Research on Agent-Based and Generative Music Recommendation — *Beijing*

Projects entrusted by enterprises and institutions with *ByteDance*, PI (Prof. Zhao) — Apr. 2026 - Now

- Generative Recommendation
- Agentic RecSys

Research on Generative Auto-Bidding — *Online*

Projects entrusted by enterprises and institutions with *Alibaba*, PI (Prof. Li) — Oct. 2025 - Now

- DT-based Generative Auto-bidding

Research on Personalized Retrieval Augment Generation — *Shenzhen*

Projects entrusted by enterprises and institutions with *Huawei*, PI (Prof. Zhao) — Apr. 2025 - Apr.2026

- Cloud-Device SLM LLM Collaboration
- Personalized Pre-Retrieval
- Personalized Retrieval

Research on Cross Business Domain Complementary and Fusion Methods from Perspective of Consumer Behavior Understanding — *Beijing*

Projects entrusted by enterprises and institutions with *Meituan*, PI (Prof. Li) — Sep. 2021 - July. 2022

- Consumer Behavior and Intention Modeling
- Milti-Task and Multi-Domain Recommendation